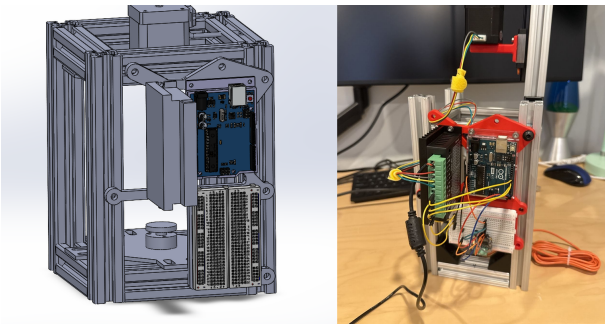




Purpose-built cyclic fatigue testing for compliant mechanisms

Closed-loop force-controlled test rig designed to validate the fatigue life of topology-optimized flexures generated by the deFlex platform. Records complete force-displacement curves each cycle, enabling detection of stiffness degradation, hysteresis changes, and nonlinear response characterization.

Built for under \$1,000. Self-reacting C-frame requires no external anchoring. Over 100,000 cycles per day.

Key Specifications

Max force	196 N (20 kg)	Sample rate	80 Hz, synced with position
Force resolution	24-bit ADC, sub-0.01 N	Cycle rate	>100,000 cycles/day
Force accuracy	±0.2% of rated capacity	Control mode	Closed-loop force-controlled
Stroke	50.8 mm total travel	Data output	CSV per cycle
Position resolution	0.008 mm	Actuator	Captive linear stepper
Working envelope	≈ 4" × 4" × 4"	Frame	80/20 aluminum C-frame

Test Modes

Force-controlled cyclic fatigue — compresses in displacement increments, reads force at each step, reverses at target force. Full force-displacement curve per cycle.

Also supports displacement-controlled cycling, static characterization, and ramp-to-failure.

Specimen Types

Topology-optimized compliant mechanisms from deFlex, conventional flexures (leaf springs, notch hinges, blade flexures), and compliant valve mechanisms.

3D-printed polymers and metals (FDM, SLA, SLS) or machined aluminum, steel, and titanium.

Validation Results

Metric	Measured	Rated (2006N397)	Status
Spring rate	9.65 ± 0.07 N/mm	10.68 ± 1.07 N/mm	Within tolerance
Linearity	$R^2 > 0.9999$	Linear	Confirmed
Cycle variation	< 0.07 N/mm	—	Repeatable
Deviation	9.6%	±10% rated tolerance	Within spec

Validated against 302 stainless steel compression spring across 10 complete load-unload cycles. Measured stiffness falls within the manufacturer’s rated tolerance band. Cycle-to-cycle variation is negligible.